

Probabilistic Sensing, Deterministic Authority: Admitting Model-Produced Observations into Sufficiency-Checked Governance Contracts

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Abstract

When a field that an authority contract needs exists only in unstructured evidence, a model can sense it. We admit the model’s output only as an observation record with a score. An admission policy, with thresholds fitted on a held-out split at a declared false-positive ceiling, maps each score to true, false or unknown. Unknown denies. A deterministic, sufficiency-checked contract decides. The probability that sensing changes the verdict is bounded by the sum, over the contract’s sensed fields, of the admitted-wrong and unknown rates. This is an instantiation of union-bound reasoning, indexed by the contract. Minimising the estimated bound is a valid cost model for choosing among sufficient contracts. In a registered study on two constructed domains with two sensor families (36,000 model calls), no cell refuted the bound. Deny-to-allow flips appeared for the first time in this programme: 13 of 21,000 test verdicts, all from 3 contradictory records, each under its bound. Sensing-aware selection picked the lower-exposure contract in 4 of 4 registered tests. Both sensors’ scores were informative but not calibrated. Correctness is relative to the declared loss model, candidate representation and reachable states; all domains are constructed.

1 Introduction

Papers 1 to 5 of this series govern agent actions with facts read from systems of record. Paper 5 compiles, from a declared loss model and a declared reachable-state model, the observation contracts that are sufficient to reproduce a verdict, and selects among them by cardinality or declared cost [5]. Every field those contracts read is recorded: it comes from a system that stores it, with metadata a data-quality check can test. That leaves a gap. Some fields an authority contract needs are written down only in unstructured evidence: a change ticket that says an approval was given, a contract that says where data are stored.

What changes. This paper adds a third kind of input. A model may read the document and *sense* the field. The sensor writes an observation record with a score per candidate value. An admission policy, a pair of thresholds per value fitted on held-out data, maps the score to true, false or unknown. The contract then decides exactly as before.

What does not change. The gate code, the compiler and the invariants of the earlier papers stay as they are. A sensed record is a new record type in the governed buffer; it is never written back to a source system. The invariant this paper adds is one sentence. A model’s output enters the gate as an observation with an admission policy. It never carries authority.

Contributions.

- *Architecture (claim C1).* A sensed field enters the gate only as a stamped observation record with provenance, score and admission decision. The model never emits a verdict (Sections 2 and 3).

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- *Bound and selection (claims C2 and C3).* Proposition S1 bounds the probability that sensing changes the verdict by the sum, over the contract’s sensed fields, of the rates of wrong admission and of unknown. Proposition S2 makes the estimated bound a valid cost model for paper 5’s selection, for nested sensed-field sets (Section 4).
- *Registered evidence (claims C4, C5 and C7).* On two constructed domains, with two sensor families and three noise levels, no registered cell refuted the bound; the first deny-to-allow flips of the programme appeared, rarely and each under its bound; and sensing-aware selection picked the contract with lower measured exposure in every registered test (Section 7).
- *Calibration (claim C6).* Neither sensor family’s scores were calibrated, although both were informative. Thresholds must therefore be fitted, not assumed (Section 7).

Reader’s map. Section 2 sets out the objects and Figure 1 the pipeline. Section 3 states the invariants. Section 4 gives the propositions and their proof status. Section 5 describes admission and estimation. Section 6 summarises the registered design and Section 7 the results. Sections 8 to 11 cover related work, claims, limitations and reproducibility. Appendix A carries the proofs and Appendix B the verbatim registered materials.

2 Setting

We inherit paper 5’s notation. A is the set of candidate attributes, R the set of reachable tuples, and $g : R \rightarrow \{\text{allow}, \text{deny}\}$ the declared verdict. A contract $C \subseteq A$ is *sufficient* if g is determined by C on R : two reachable tuples that agree on C get the same verdict.

New objects. The contract is partitioned as $C = F_r \cup F_s$. Recorded fields F_r are read from systems of record. Sensed fields F_s are produced by sensors. A sensor S_i for field i maps an evidence document d to a score $s \in [0, 1]$ for each candidate value of the field, one value against the rest. An admission policy A_i maps each value’s score to true, false or unknown with two thresholds, τ^{true} and τ^{false} , fitted at a false-positive target ε . The field’s admitted value is v if exactly one value maps to true; otherwise the field is unknown, and unknown denies.

The sensed record. Each admission is written as a record with the fields `request_id`, `resource_id`, `field`, `value`, `score`, `admission`, `admission_policy_version`, the sensor’s identifier and version, the SHA-256 of the source document and `sensed_at`. It carries metadata, so the data-quality predicates of the earlier papers apply to it: freshness of sensing, the sensor version pin, the admission policy version pin, and lineage to the document, the request and the resource. A record whose stamp does not match the request being decided is rejected and the field is unknown.

Evaluation from observations. Given an observed value for every field of C (recorded values as recorded, sensed values as admitted, derived values computed as in Section 3), the verdict is $g(t')$ for any $t' \in R$ that agrees with the observations on C . Sufficiency makes this well defined. If a sensed field is unknown, or no reachable tuple agrees, the verdict is deny.

Exposure. For a request with true tuple t , exposure is the event that the verdict from recorded and sensed values differs from the verdict from recorded and true values. A change from deny to allow is *unsafe*; a change from allow to deny is *fail-closed*. For sensed field i and a distribution over requests, e_i is the probability that the field is admitted with a wrong value and u_i the probability that it is unknown.

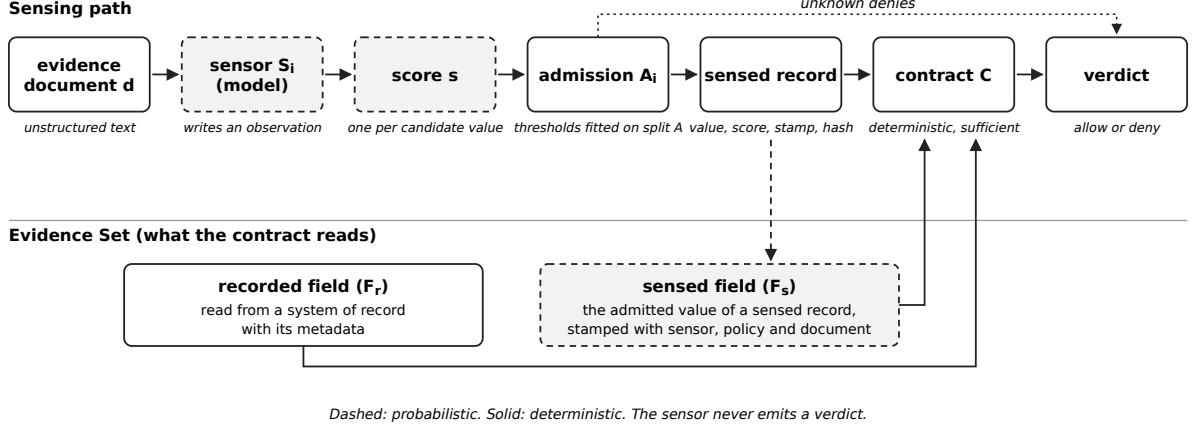


Figure 1: The sensing pipeline. The top row is the sensing path from document to verdict; dashed boxes are probabilistic and solid boxes deterministic. The bottom row is the Evidence Set line: the contract reads a recorded field and a sensed field side by side, and the sensed field reaches it only as the admitted value of a stamped record.

3 Invariants

I1, carried from paper 3. Validity is never elicited from a model. The sensor writes an observation; the deterministic contract computes the verdict. The admission policy is a threshold with abstention per field, never a weighted combination across fields.

This paper makes one exception to the earlier practice of reading every field from a record, and states it precisely. A sensed field’s value may come from a model, but only as an admitted *assertion*: what the document states. Nothing else about the field comes from the model.

I2. A sensor is asked only what the document asserts: whether an approval is recorded or not recorded, the date the document gives, where it says the data are stored, and which branch or environment it names. Validity, scope and expiry are never sensed. They are computed deterministically from recorded fields plus the admitted assertion. On the CH-C1 domain, for example, the sensed field is `approval_assertion`, and `approval_token` is derived from it: absent if the assertion is `not_recorded`; valid if the asserted date lies within 30 days before the recorded request date; expired otherwise.

I2 answers the obvious objection to letting a model near an authority gate. The model is not asked whether an approval is valid, current or sufficient. It is asked what a document says, which is a reading task with a checkable label. The judgement that turns the reading into a permission stays in code that can be inspected and tested.

4 Propositions

Proposition S1 (union bound). For a contract with sensed-field set F_s ,

$$P(\text{verdict change}) \leq \sum_{i \in F_s} (e_i + u_i), \quad P(\text{deny} \rightarrow \text{allow}) \leq \sum_{i \in F_s} e_i.$$

Sketch. If every sensed field is admitted with its true value, the observations agree with the true tuple on C , so sufficiency gives the true verdict. A verdict change therefore requires at least one sensed field that is wrong or unknown. A change to allow also requires that no field is unknown, because unknown denies, so it requires at least one wrongly admitted field. Both inequalities follow from the union bound. The argument is pointwise: the event inclusion holds at every pair of a true tuple and an observation vector, and both sides are linear in the joint

distribution, so the inequalities hold for every distribution, including any dependence between fields. S1 is an instantiation of union-bound reasoning over program events [4], indexed by the contract’s sensed fields. It is not a new inequality.

Proposition S2 (monotonicity for nested sets). For two sufficient contracts whose sensed-field sets are nested, $F'_s \subseteq F_s$, read by the same sensors on the same documents, the S1 bound of F'_s does not exceed that of F_s . The selection objective is the *estimated* bound $\sum_{i \in F_s} (\hat{e}_i + \hat{u}_i)$ of Section 5. It fits paper 5’s observation-cost interface, so no new solver is needed. No claim is made for sets that are not nested. The S2 checker records why: with the per-field rates below, a rule that prefers fewer sensed fields picks the one-field set, while the registered objective picks the two-field set, whose bound is a hundred times lower.

sensed-field set	fields	e per field	S1 bound	picked by
one field	a	0.2, 0.0	0.2	fewest sensed fields
two fields	b, c	0.001, 0.0; 0.001, 0.0	0.002	registered objective

The rate columns list e and then u for each field, as the checker records them.

Proposition S3 (fail-closed). Call a sensing configuration *deny-ward* if, for every reachable tuple t with $g(t) = \text{deny}$ and every observation vector that agrees with t on F_r and whose sensed fields are each either true or admitted with any wrong value, jointly, the contract evaluates to deny. If unknown routes to deny and the configuration is deny-ward, then deny-to-allow changes have probability zero. The condition is global over R and over joint errors, not field by field.

N6 (negative witnesses). Two constructions mark the edges of S1 and S3. In witness (a), two binary fields with disjoint wrong admissions attain the S1 deny-to-allow bound exactly. In witness (b), an AND contract with true values (false, false) meets a sensor that admits both fields wrongly together with probability q and never one alone. Every single-field error is harmless, so a field-by-field direction test calls the configuration deny-ward, yet the deny-to-allow probability is q . The first registration of this paper used such a directional bound; witness (b) defeats it. The external review of the novelty fence raised the point, and the re-registration from **prereg-p6-v1** to **prereg-p6-v1.1** replaced the directional term with S1 and made S3’s condition global.

Each statement carries the proof-status tag recorded in **proof_status.json**. A tag of machine-checked means a checker holds on exactly the statement’s scope; checked-scope-only means a checker holds on a narrower scope; pending-human-review means no checker covers it.

statement	scope	checker	tag
S1 on the finite model families	finite-models:B12,B3,T2	s1_check	machine-checked
S1, general	general	s1_check	checked-scope-only
S2 on the finite model families, nested sets	finite-models:B12,B3,T2;nested-sets-only	s2_check	machine-checked
S2, general	general	s2_check	checked-scope-only
S3 on the finite model families	finite-models:B12,B3,T2	s3_check	machine-checked
S3, general	general	s3_check	checked-scope-only
N6, both witnesses	witness	n6_witness	machine-checked
written proof of S1 (Appendix A)	general	none	pending-human-review
written proof of S2 (Appendix A)	general	none	pending-human-review
written proof of S3 (Appendix A)	general	none	pending-human-review

5 Admission policy and estimation

Each experiment’s validation split is divided once, before any model call, into split A and split B of 150 items each (seed 20261107). The two halves have different jobs.

Split A fits the thresholds against the assertion label. For each sensor, field and candidate value v , τ_v^{true} is the lowest candidate threshold at which at most ε of split-A records whose assertion label for v is zero score at or above it; τ_v^{false} is the highest at which at most ε of records whose label is one score at or below it. The assertion label says what the document states, so the thresholds measure reading fidelity. The ceiling ε is 1%, with 5% as an ablation. The ceiling is a fitting target on split A, not a property of the test split.

Split B estimates the error rates against the truth label. With the thresholds frozen, $\hat{e}_i$ and $\hat{u}_i$ are one-sided 95% Clopper-Pearson upper bounds [8] on the split-B rates of wrong admission and of unknown, per sensor, field and noise level. The truth label is the item’s true value. A faithful reading of a contradictory document can therefore count as a wrong admission. The registered bounds per cell are $B^+ = \sum_{i \in F_s} \hat{e}_i$ for deny to allow and $B = \sum_{i \in F_s} (\hat{e}_i + \hat{u}_i)$ for any change. Thresholds and estimates were written to a committed file before any test-split call.

The floor. A Clopper-Pearson upper bound is never zero. With no error among 150 split-B records, the bound for one field is 0.0198. With two sensed fields, a sensor that made no split-B error still gets $B^+ = 0.0395$ and $B = 0.0791$. The bounds are therefore valid, not tight. Section 7 shows how far apart the two are.

6 Experimental design

Everything below was registered before any experiment code existed. The first registration is `prereg/p6-v1.md` (tag `prereg-p6-v1`, commit `ddcddf4`). After an external review of the novelty fence, the author re-registered as `prereg/p6-v1.1.md` (tag `prereg-p6-v1.1`, commit `2fac5a4`). The review is committed unedited under `review-secondary/` and was not applied to any file.

Sensors. Two sensor families read the same documents. Jev (System One, model `jev-latest`, returned as `jev-1.13.0`) is a typed probabilistic model from TypeSafe; it receives one question per candidate value. Claude Haiku 4.5 (`claude-haiku-4-5-20251001`) is a generative language model from Anthropic; it receives one prompt per field and returns a JSON object of probabilities. TypeSafe and Anthropic appear here only as the vendors of the tested configurations. The questions and the prompt are in Appendix B.

E0, motivating evidence. The earlier probes `jev-v1`, `jev-v2` and `jev-v3` are quoted from tag `jev-probes-final` (commit `e1fc75b`). Nothing is re-run.

E1, the bound on CH-C1. Paper 5’s 35-property domain, with contract K, the first minimum-cardinality contract of paper 5’s CH-C1 checker. A registered rule picks the two document-backed fields of K with the highest declared observation cost: `approval_token`, sensed as `approval_assertion` over `not_recorded` and six registered dates, and `data_residency_region`. All other fields are recorded at their true values. The population is 27,000 reachable tuples. From it, 1,000 items were drawn (seed 20261101), with a validation split of 300 (seed 20261102) and a test split of 700.

E2, the substitution test on CH-B1. Paper 5’s code and cloud domain, with `jev-v2`’s items and split. CH-B1 has two seven-attribute reducts, R_{branch} and R_{env} . In arm 1, `branch` is sensed and `environment` recorded, so R_{env} senses only the approval field and R_{branch} senses two fields. Arm 2 is the mirror image. In each arm the sensed-field sets are nested, so S2 applies. The registered rule picks the reduct with the lower estimated bound B on split B at 30% noise, before any test-split call. Both reducts are evaluated on the same readings, so every test item gives a pair of outcomes.

E3 to E5. E3 re-reads the E1 caches at ceilings of 1% and 5%, and with unknown routed to deny or to escalation. E4 runs the two N6 witnesses through the pipeline on 10,000 synthetic items each, with no model. E5 measures calibration against the assertion label.

Documents and labels. Each item gets one synthetic record per noise level, generated from a seeded generator and never edited. The record carries no property value except the sensed statements. Validation and test items draw their statements from disjoint phrase banks. At noise levels 0%, 10% and 30%, each sensed statement is independently perturbed: *missing* removes it, and *contradictory* keeps it and appends a comment asserting a different value. Noise is nested across levels. Each item and field has two labels. The *truth label* is the true value. The *assertion label* marks every value the record states, so a contradictory record asserts two values and a missing one asserts none.

Run discipline. The order was registered: E1 before E2; within each, all validation records, then the arm health check, then thresholds written and committed, then the test split. The arm health rule stops a run before any further call if more than 2 invalid replies occur in the first 20 validation records of a sensor, or if more than 5% of its validation replies are invalid. A cap of USD 60 covered E1 to E5. Every raw response was appended to a cache before parsing. Every number in the results is a named slot filled from the caches by one command.

7 Results

Runs. Both experiments completed: 36,000 calls (12,000 in E1 and 24,000 in E2), 0 invalid replies, USD 8.26 in total. Returned model strings: claude-haiku-4-5-20251001, jev-1.13.0 (E1); claude-haiku-4-5-20251001, jev-1.13.0 (E2).

7.1 E0: the earlier probes

The earlier probes are quoted from tag `jev-probes-final`. Across two sensors, three noise levels and 4,200 test verdicts on one sensed field, no sensing error turned deny into allow (0 of 4,200); every change was fail-closed. Calibration was refuted for Jev in jev-v1 (Spiegelhalter $Z = -15.0268$, Holm-adjusted $p = 1.47e-50$, verdict: refuted). Known issue KI-1 is part of the record. In jev-v2, Haiku wrapped every reply in a code fence and the registered parser scored 3000 of 3000 replies invalid, so that arm denied everything. jev-v3 re-ran the arm under a registered fix and an arm health check, with 0 invalid replies. The arm health rule of this paper comes from that record.

7.2 E1: the bound on CH-C1

Each cell is a sensor at a noise level, on the 700 test items. Rates carry Clopper-Pearson 95% intervals. B^+ and B are the split-B bounds of Section 5.

sensor	noise	unsafe (k/n)	unsafe 95% CI	B^+	change rate (k/n)	change 95% CI	B	no unsafe change
Jev	0%	0/700	[0.0000, 0.0053]	0.0395	0.0000 (0/700)	[0.0000, 0.0053]	0.0791	yes
Jev	10%	0/700	[0.0000, 0.0053]	0.0611	0.0429 (30/700)	[0.0291, 0.0606]	0.3045	yes
Jev	30%	1/700	[0.0000, 0.0079]	0.1056	0.1071 (75/700)	[0.0852, 0.1324]	0.7837	no
Haiku	0%	0/700	[0.0000, 0.0053]	0.0395	0.0643 (45/700)	[0.0473, 0.0851]	0.4268	yes
Haiku	10%	0/700	[0.0000, 0.0053]	0.0798	0.0943 (66/700)	[0.0737, 0.1184]	0.6324	yes

sensor	noise	unsafe (k/n)	unsafe 95% CI	B^+	change rate (k/n)	change 95% CI	B	no unsafe change
Haiku	30%	0/700	[0.0000, 0.0053]	0.0707	0.1471 (103/700)	[0.1217, 0.1756]	1.0761 vacuous	yes

A bound greater than one is marked vacuous: it holds for any sensor. Vacuous bounds in E1: B for Haiku at 30% noise. The floor of Section 5 applies to every cell: with no split-B error among 150 records per field, the bound is 0.0198 for `approval_assertion` and 0.0198 for `data_residency_region`, so B^+ cannot fall below 0.0395 and B below 0.0791.

No cell’s unsafe rate exceeded its B^+ (H1, Section 7.8). The bounds are valid, not tight. Every B^+ is at least 0.0395, while the largest unsafe rate observed in any cell is 0.0014. Where verdicts changed at all, B was 6.6 to 7.3 times the observed change rate.

Per field, on the test split, the rates of wrong admission and of unknown were:

sensor	noise	approval: wrong	approval: unknown	residency: wrong	residency: unknown
Jev	0%	0.0000	0.0000	0.0000	0.0000
Jev	10%	0.0114	0.0671	0.0100	0.0929
Jev	30%	0.0514	0.1714	0.0186	0.2700
Haiku	0%	0.0000	0.2443	0.0000	0.0000
Haiku	10%	0.0171	0.2943	0.0029	0.1086
Haiku	30%	0.0400	0.4014	0.0100	0.3071

Haiku’s unknown rate on the approval field is high even without noise. The thresholds explain it. On split A, no candidate threshold for Haiku’s `not_recorded` value met the 1% ceiling in either direction: its fitted τ^{true} is “never true” and its τ^{false} is “never false” (Appendix B). That value can never be admitted, so Haiku cannot read a record that asserts no approval correctly: the field is unknown, or wrong if a date value is admitted instead. At 0% noise its wrong rate on this field was 0.0000, so the admission policy turned a poorly separated score into abstention, and abstention denies.

7.3 The first flips

This programme had recorded no deny-to-allow change before this paper. E1 and E2 together recorded 13 of 21,000 test verdicts: 1 of 4,200 in E1 (Jev at 30% noise) and 12 of 16,800 in E2. No cell had more than 1 flip in 700, and every flip lay under its cell’s registered bound.

The count overstates the number of independent events. The item-level record (`results/p6-flips.json`, written from the caches by `python -m experiments.flips_p6`) traces all 13 flips to 3 records: item 6042 in E1, item 1595 in E2 arm 1 and item 2312 in E2 arm 2. Every flip was on a record whose approval statement had been made contradictory (all on contradictory approval statement: yes), and in every flip the approval field was the only wrongly admitted field (yes). In E2, both sensors flipped on each such record (yes), under both reducts (yes), because both reducts read the same approval field. The E1 record shows the mechanism. Its true approval had been recorded but had expired at the request date, so the true verdict was deny. The appended comment said the approval log was empty. Jev admitted `not_recorded`, the derived token became absent, and contract K allows that combination.

Interpretation. The flips are not misreadings of a clear document. Each record stated two incompatible things about the approval, and the sensors admitted the statement in the appended comment. The truth label counts that as a wrong admission, as registered: exposure is measured against what is true, not against what the document says. The paired flips in E2 are one event per record, seen through two reducts and two sensors, not 12 independent failures.

Both reducts read the same approval field, so neither choice of reduct can avoid this failure mode.

7.4 E2: substitution test on CH-B1

The picks were fixed on split B at 30% noise, before any test-split call. In both arms and for both sensors, the rule picked the reduct that senses only the approval field.

sensor	arm	estimated bound, R_{branch}	estimated bound, R_{env}	picked
Jev	arm 1	0.7504	0.3171	R_env
Jev	arm 2	0.3170	0.7157	R_branch
Haiku	arm 1	0.9182	0.4837	R_env
Haiku	arm 2	0.5476	0.9884	R_branch

On the test split, the verdict-change and unsafe rates were:

sensor	arm	noise	picked: change	other: change	picked: unsafe	other: unsafe
Jev	arm 1	0%	0.0000 (0)	0.0000 (0)	0	0
Jev	arm 1	10%	0.0286 (20)	0.0514 (36)	0	0
Jev	arm 1	30%	0.0943 (66)	0.1943 (136)	1	1
Jev	arm 2	0%	0.0000 (0)	0.0000 (0)	0	0
Jev	arm 2	10%	0.0343 (24)	0.0714 (50)	1	1
Jev	arm 2	30%	0.0757 (53)	0.1786 (125)	1	1
Haiku	arm 1	0%	0.1443 (101)	0.1443 (101)	0	0
Haiku	arm 1	10%	0.1586 (111)	0.1714 (120)	0	0
Haiku	arm 1	30%	0.1714 (120)	0.2371 (166)	1	1
Haiku	arm 2	0%	0.1457 (102)	0.1657 (116)	0	0
Haiku	arm 2	10%	0.1443 (101)	0.1829 (128)	1	1
Haiku	arm 2	30%	0.1471 (103)	0.2214 (155)	1	1

Counts are out of 700 test items per cell. At 30% noise the picked reduct changed fewer verdicts than the other in all four registered comparisons (H2, Section 7.8). The unsafe columns are equal in every row, because every flip came from the shared approval field. E2 is a substitution test between two reducts with nested sensed-field sets. It is not a general selection experiment.

7.5 E3: admission ablation

E3 re-reads the E1 caches; no new call was made. Coverage is the fraction of test items decided. The unsafe rate is over all test items. Exposure is the verdict-change rate among decided items.

ceiling	unknown	sensor	noise	coverage	unsafe rate	exposure among decided
1%	deny	Jev	0%	1.0000	0.0000	0.0000
1%	deny	Jev	10%	1.0000	0.0000	0.0429
1%	deny	Jev	30%	1.0000	0.0014	0.1071
1%	deny	Haiku	0%	1.0000	0.0000	0.0643
1%	deny	Haiku	10%	1.0000	0.0000	0.0943
1%	deny	Haiku	30%	1.0000	0.0000	0.1471
1%	escalate	Jev	0%	1.0000	0.0000	0.0000
1%	escalate	Jev	10%	0.8457	0.0000	0.0034
1%	escalate	Jev	30%	0.6000	0.0014	0.0167
1%	escalate	Haiku	0%	0.7557	0.0000	0.0000

ceiling	unknown	sensor	noise	coverage	unsafe rate	exposure among decided
1%	escalate	Haiku	10%	0.6243	0.0000	0.0000
1%	escalate	Haiku	30%	0.4086	0.0000	0.0175
5%	deny	Jev	0%	1.0000	0.0000	0.0000
5%	deny	Jev	10%	1.0000	0.0000	0.0286
5%	deny	Jev	30%	1.0000	0.0014	0.0914
5%	deny	Haiku	0%	1.0000	0.0000	0.0429
5%	deny	Haiku	10%	1.0000	0.0000	0.0643
5%	deny	Haiku	30%	1.0000	0.0000	0.1271
5%	escalate	Jev	0%	1.0000	0.0000	0.0000
5%	escalate	Jev	10%	0.8843	0.0000	0.0048
5%	escalate	Jev	30%	0.6700	0.0014	0.0235
5%	escalate	Haiku	0%	0.8457	0.0000	0.0000
5%	escalate	Haiku	10%	0.7200	0.0000	0.0020
5%	escalate	Haiku	30%	0.5029	0.0000	0.0256

Routing unknown to escalation instead of deny trades coverage for exposure among decided items: fewer items are decided automatically, and in every cell where any verdict changed, the decided items changed verdict less often. Raising the ceiling from 1% to 5% raised coverage under escalation in every cell where it was below one. The unsafe rate did not move between the two ceilings. These readings are descriptive; E3 registered no hypothesis.

7.6 E4: witness and implementation check

E4 runs the two N6 witnesses through the experiment pipeline, with no model.

witness	measured deny-to-allow (95% CI)	checker's exact rate	S1 bound	pipeline matches checker
(a) disjoint errors	0.0452 [0.0412, 0.0495] (452/10000)	1/20	1/20	yes
(b) joint errors, AND contract	0.0503 [0.0461, 0.0548] (503/10000)	1/20	1/10	yes

The registered check is that the pipeline's per-item outcomes equal the checker's own simulation, and it passed for both witnesses (check passes: yes). In witness (b) the S1 bound is twice the exact rate and holds; the v1 directional bound, 0, lies below the exact rate and fails. One reading needs care. In witness (a) the exact rate lies just above the simulation's 95% interval. That is a property of one seeded draw of 10,000 items, not of the pipeline: the checker's own simulation, with the same seed, gives the same 452 flips.

7.7 E5: calibration

Calibration is measured on the test-split scores of every field and value question, against the assertion label.

domain	sensor	answers	Spiegelhalter Z	p	Brier	BSS
CH-C1	Jev	21000	-40.5181	0.00e+00	0.0065	0.9566
CH-C1	Haiku	21000	7.1036	1.22e-12	0.0142	0.9058
CH-B1	Jev	21000	-48.8080	0.00e+00	0.0113	0.9516
CH-B1	Haiku	21000	-3.6214	0.0003	0.0315	0.8655

Both sensor families were informative, not calibrated. The Brier skill score [6] ranged from 0.8655 to 0.9566, far above zero, the score of a forecast that always gives the base rate. The Spiegelhalter test [16] rejected calibration in all four cases (largest $p = 0.0003$). A high score therefore does not mean an equally high chance that the document says so [10]. This is why the thresholds are fitted on held-out data and not read off the score.

7.8 Hypotheses

The registered family is H1 and the four H2 tests, with Holm’s correction [11] at $\alpha = 0.05$. H1 combines the 6 E1 cells’ exact one-sided binomial tests with a Bonferroni factor. H2 uses an exact McNemar test [13] on the paired outcomes at 30% noise. Intervals for H2 come from an item-clustered bootstrap with 10000 resamples (seed 20261104). The table is as in `results/p6.md`.

hypothesis	estimate	95% CI	p	Holm-adjusted p	verdict
H1: no E1 cell’s unsafe rate exceeds its split-B bound (Bonferroni over 6 cells)	see E1 table		1.0000	1.0000	not refuted: no violation detected
H2, Jev, arm 1: picked reduct has lower exposure at 30% noise	-0.1000 (b = 0, c = 70, n = 700)	[-0.1229, -0.0786]	1.69e-21	6.78e-21	supported: picked reduct lower
H2, Jev, arm 2: picked reduct has lower exposure at 30% noise	-0.1029 (b = 0, c = 72, n = 700)	[-0.1257, -0.0814]	4.24e-22	2.12e-21	supported: picked reduct lower
H2, Claude Haiku 4.5, arm 1: picked reduct has lower exposure at 30% noise	-0.0657 (b = 0, c = 46, n = 700)	[-0.0843, -0.0486]	2.84e-14	5.68e-14	supported: picked reduct lower
H2, Claude Haiku 4.5, arm 2: picked reduct has lower exposure at 30% noise	-0.0743 (b = 0, c = 52, n = 700)	[-0.0943, -0.0557]	4.44e-16	1.33e-15	supported: picked reduct lower

“Not refuted” for H1 means no violation was detected on 700 items per cell. It is never presented as evidence that the bound holds in general. Everything outside this table is descriptive.

8 Related work and fence

Not claimed as new. Thresholding a score with a reject option goes back to Chow [7] and is studied for deep networks as selective classification [9]. Conformal prediction and risk control give distribution-free coverage for set-valued predictions [1, 17]. Learning to defer trains a model to hand cases to a human [12, 14]. Calibration measurement is standard [6, 10, 16]. Confidence-scored information extraction and human-in-the-loop escalation are common practice. Runtime enforcement and shields were fenced in paper 4, and reduct theory and cost-sensitive reduction [15] in paper 5.

Prior compositions. Composing an uncertain sensing component with a symbolic check that decides has been done before. Zhu and Zhang add probabilistic attributes, inferred by learned models, to access policies with automated reasoning support; their location example

obtains an access-control attribute through probabilistic sensing [19]. Astorga et al. synthesise perception contracts and check that they preserve a downstream invariant [3]. Artikis et al. threshold uncertain observations and process the admitted events deterministically [2]. The composition of a sensor with a deciding check is therefore not claimed as new.

Union bounds. Barthe et al. give a program logic whose judgements carry a failure budget that adds under composition [4]. Assigning each sensed field the assertion “correct and definite” and using sufficiency to imply verdict agreement instantiates that reasoning; S1 restricts the sum to the contract’s sensed fields. Waite et al. combine perception-error union bounds with deterministic verification of autonomous systems [18]. Neither states a bound indexed by an authority contract’s sensed fields, but both occupy the broader statistical-to-symbolic composition.

What is claimed is the conjunction: an abstaining admission policy between a sensor and a sufficiency-checked authority contract; an exposure bound stated over that contract’s sensed fields; and sensing-aware selection among sufficient contracts as a paper 5 cost model, for nested sensed-field sets. Neither the admission policy alone, nor the bound alone, nor the composition with a symbolic decision layer alone is claimed. This paper does not introduce abstention, calibration, conformal control or reducts. It shows how an abstaining sensor composes with a sufficiency-checked contract, bounds the resulting exposure by the contract’s sensed fields, and selects contracts to minimise it.

9 Claims versus non-claims

Each claim’s status is taken from `CLAIMS.md`, where the evidence column names the slot or checker output behind it.

claim	statement	type	status
C1	A sensed field enters the gate only as an observation record with provenance, score and admission stamp; the model never emits a verdict.	architecture	supported
C2	Proposition S1 bounds the probability of a verdict change by the sum of the sensed fields’ wrong and unknown rates, and of a deny-to-allow change by the sum of their wrong rates.	instantiation	instantiation
C3	For nested sensed-field sets, the bound does not increase as sensed fields are removed; minimising the estimated bound is a paper 5 cost model.	formal	supported (finite models, nested sets)
C4	On CH-B1 and CH-C1, with two sensor families and three noise levels, the observed unsafe rate never exceeds the bound.	empirical, registered	not refuted
C5	Deny-to-allow flips were rare (13 of 21,000 test verdicts, each 1 in 700) and every one lay under its registered bound.	empirical, registered	supported
C6	Scores from both sensor families are not calibrated, so admission thresholds must be set empirically.	empirical	supported
C7	Sensing-aware reduct selection changes which reduct is chosen on CH-B1 and reduces measured exposure.	empirical, registered	supported
C8	Everything reproduces from committed caches; no number is hand-entered.	reproducibility	supported

The non-claims, verbatim from `CLAIMS.md`: “no prevalence in live systems; no claim that sensing is safe in general; no claim about Jev or Haiku beyond the tested configurations; no claim that the bound is tight.”

Correctness is relative to the declared loss model, candidate representation and reachable states; all domains are constructed.

10 Limitations

Constructed corpora. All documents are synthetic, in templated English, one statement per sensed field. Validation and test phrase banks are disjoint, but the rest of the document is shared across splits: the header lines, the filler bank, the noise templates, the date set, the names and the layout. A sensor can benefit from surface regularities that carry over from validation to test. No prevalence claim is made.

Loose bounds. The bounds are valid, not tight. Every B^+ in E1 is at least 0.0395, while the largest observed unsafe rate is 0.0014. Where verdicts changed, B was 6.6 to 7.3 times the observed change rate. Three causes add up. The Clopper-Pearson floor charges every sensed field a cost even with no observed error. The union bound adds per-field terms that overlap in practice. And the truth label counts every faithful reading of a contradictory record as a wrong admission, although most such readings do not change the verdict.

Two sensor configurations. One configuration of each sensor family was tested, through the vendors' interfaces as pinned, on the run dates recorded in the manifests. Nothing is claimed about other versions, prompts or models.

Shared-field dependence in E2. Both reducts of an arm read the same approval field. Their outcomes are paired and their flips coincide (Section 7.3). The paired test accounts for this; the count of flips does not measure independent events.

Stamp enforcement. Record binding and stamp enforcement (a record whose request, resource, sensor version or policy version does not match is rejected) are covered by unit tests only. No experiment presented a mismatched record.

Proof status. The general proofs in Appendix A are tagged pending-human-review. The checkers cover finite model families and the witnesses.

11 Reproducibility

Siblings and pins. `engines.lock` pins `sarc-authority-derivation` at `cfb321e` (CH-B1 and CH-C1), `dqSarc` at `db6c396`, and this repository's own Jev probe record at `e1fc75b` (tag `jev-probes-final`). A run aborts before any model call if a pin differs.

Caches and slots. The raw responses are committed append-only in `responses/p6-E1.jsonl` and `responses/p6-E2.jsonl`, with run manifests. `python -m experiments.analysis_p6` reads only the caches, the frozen thresholds and the pinned siblings, and writes `results/p6.md` and `results/p6.slots.json`; two runs give identical bytes. This manuscript is filled from those slots, the checker outputs and the E0 slots at the pinned tag by `python paper/populate.py`; a lint rejects any typed numeral that is not a registered design parameter. A reproduction from the caches calls no model.

Checkers and mutation testing. `make formal` runs the S1, S2, S3 and N6 checkers twice and requires identical output. The S1 checker evaluated 4400, 131072 and 115200 pointwise cases on the three finite model families. Mutation testing of `src/sensed_authority` with `mutmut 3.7.0` killed 489 of 511 mutants with 0 untested, a kill score of 0.957 against a threshold of 0.85.

Review and re-registration. The external fence review, its attestation, the superseded registration and the re-registration are all committed and tagged. Departures are logged in `prereg/DEVIATIONS.md`; none was recorded for this paper.

Standing policy. The arm health rule, introduced after KI-1, is now standing policy for every registered model run in this programme: a smoke check on the first 20 validation records and a 5% invalid ceiling on the validation split, both final.

Build. `make paper` regenerates the manuscript, the figure, the bibliography and the PDF. `make release-check` runs the tests, the checkers, the analysis regeneration, the lints and the release gates.

Acknowledgements

An external model review of the novelty fence was commissioned before registration, committed unedited and not applied; the author re-registered after reading it. An independent reproducer will be named in a later version, with permission. AI assistance was used for drafting and engineering, under the author’s direction; the author is solely responsible for the content.

A Proofs and checker provenance

Proof of S1. Fix a request with true tuple t and let o be the observation vector: recorded fields at their recorded values, sensed fields at their admitted values or unknown, derived fields computed from these. Let W_i be the event that sensed field i is admitted with a wrong value and U_i the event that it is unknown. Suppose no W_i and no U_i occurs. Then o agrees with t on every field of C . Since $t \in R$ agrees with o , the verdict from observations is $g(t')$ for some t' that agrees with t on C , and sufficiency gives $g(t') = g(t)$. So the change event lies inside $\bigcup_i (W_i \cup U_i)$. If the verdict from observations is allow, no field is unknown, because unknown denies. So the deny-to-allow event lies inside $\bigcup_i W_i$. The union bound gives both inequalities. The inclusions hold at every pair (t, o) , and both sides are linear in the joint distribution of (t, o) , so the inequalities hold for every distribution, including any dependence between fields. $\square$

Proof of S2. Let $F'_s \subseteq F_s$ be nested sensed-field sets read by the same sensors on the same documents. Each field’s admitted value is the same under both contracts, so e_i and u_i are the same per field, and the readings of the smaller set are the restriction of the larger. The S1 bound is a sum of non-negative terms over the set, so removing fields cannot increase it. The estimated bound uses the same per-field split-B estimates, so it is monotone in the same way. For sets that are not nested the per-field terms differ, and Section 4 shows a counterexample to the fewest-fields rule. $\square$

Proof of S3. Suppose unknown routes to deny and the configuration is deny-ward. Take any true tuple t with $g(t) = \text{deny}$ and any realised observation. If a sensed field is unknown, the verdict is deny. Otherwise every sensed field is either true or admitted with some wrong value, and the recorded fields agree with t ; by the deny-ward condition, applied jointly to all sensed fields, the verdict is deny. So no realisation turns deny into allow, and the probability is zero. $\square$

N6 constructions. Witness (a): fields f_1, f_2 , contract allow if and only if both are true; true values are (false, true) or (true, false) with equal probability; the sensor reads the item’s false field as true with probability 0.05 and is otherwise correct, never unknown. Each field is wrong with probability $1/40$, the S1 deny-to-allow bound is $1/20$, and the exact rate is $1/20$: the bound is attained (attained: yes). Witness (b): the same contract with true values (false, false); with probability $q = 1/20$ both fields are admitted as true, otherwise both are correct. Each field is wrong with probability $1/20$; the S1 bound is $1/10$; the exact rate is $1/20$; the v1 directional bound is 0; the configuration is not deny-ward in the global sense (globally deny-ward: no).

Checker provenance. The checkers enumerate three finite model families, B12, B3 and T2. `s1_check` verified the S1 inclusions at 4400, 131072 and 115200 pairs of tuple and observation (holds: yes). `s2_check` verified S2 for nested sets at 12236, 702464 and 373248 cases (holds: yes) and recorded the counterexample of Section 4. `s3_check` enumerated 528, 2048 and 2048 configurations, of which 376, 318 and 530 were deny-ward, and confirmed that deny-ward holds exactly when no pair turns deny into allow (holds: yes). On B12 it found 4 configurations

that a field-by-field test would call deny-ward but that are not deny-ward globally. `n6_witness` computes both witnesses exactly and by simulation (holds: yes). Each output records its statement, scope, inputs hash and provenance; the provenance helper is ported with attribution from `sarc-suite-one-pass` through `sarc-authority-derivation`.

B Registered materials

Everything in this appendix is rendered from `experiments/constants_p6.py`, which the test suite checks verbatim against `prereg/p6-v1.1.md`, and from the frozen threshold files.

Jev questions. One Noul question per candidate value; each asks what the record states.

field	value	question id	question (verbatim)
approval_ assertion	not_recorded	Q_appr_none	Does this record state that no approval has been recorded for this change?
approval_ assertion	recorded	Q_appr_recorded	Does this record state that an approval for this change has been recorded?
data_ residency_ region	us	Q_residency_us	Does this record state that the resource's data is stored in the United States?
data_ residency_ region	eu	Q_residency_eu	Does this record state that the resource's data is stored in the European Union?
data_ residency_ region	apac	Q_residency_apac	Does this record state that the resource's data is stored in the Asia-Pacific region?
branch	main	Q_branch_main	Does this record state that the change targets the main branch?
branch	staging	Q_branch_staging	Does this record state that the change targets the staging branch?
branch	feature	Q_branch_feature	Does this record state that the change targets a feature branch?
environment	production	Q_env_production	Does this record state that the change targets the production environment?
environment	staging	Q_env_staging	Does this record state that the change targets the staging environment?
environment	development	Q_env_development	Does this record state that the change targets the development environment?
approval_ assertion	recorded:2026-06-01	Q_appr_20260601	Does this record state that an approval for this change was recorded on 2026-06-01?
approval_ assertion	recorded:2026-06-15	Q_appr_20260615	Does this record state that an approval for this change was recorded on 2026-06-15?
approval_ assertion	recorded:2026-07-01	Q_appr_20260701	Does this record state that an approval for this change was recorded on 2026-07-01?
approval_ assertion	recorded:2026-07-15	Q_appr_20260715	Does this record state that an approval for this change was recorded on 2026-07-15?

field	value	question id	question (verbatim)
approval_assertion	recorded:2026-08-01	Q_appr_20260801	Does this record state that an approval for this change was recorded on 2026-08-01?
approval_assertion	recorded:2026-08-15	Q_appr_20260815	Does this record state that an approval for this change was recorded on 2026-08-15?

Claude Haiku 4.5 prompt. Temperature zero, no system prompt, the assistant prefill {, and max_tokens = 64, or max_tokens = 128 for the CH-C1 approval field. The user message:

You are reading one change-management record. Report only what the record states:
→ {task}. Reply with only a JSON object of the form {object}, where each number is
→ your probability that the record states that value. No code fences. No text after
→ the object.

Record:

<rendered record text>

field	{task} (verbatim)	{object} keys
approval_assertion (CH-C1)	whether it states that no approval has been recorded, or that an approval was recorded, and on which date	p_not_recorded, p_2026_06_01, p_2026_06_15, p_2026_07_01, p_2026_07_15, p_2026_08_01, p_2026_08_15
data_residency_region (CH-C1)	where it states the resource's data are stored: us (United States), eu (European Union) or apac (Asia-Pacific)	p_us, p_eu, p_apac
approval_assertion (CH-B1)	whether it states that an approval has been recorded or that no approval has been recorded	p_recorded, p_not_recorded
branch (CH-B1)	which branch it states the change targets: main, staging or feature	p_main, p_staging, p_feature
environment (CH-B1)	which environment it states the change targets: production, staging or development	p_production, p_staging, p_development

E1 statement banks (CH-C1). {approver}, {doc} and {ref} are drawn per item; {d} is the item's approval date.

field	value	validation bank	test bank
approval_assertion	recorded	Approval: granted by {approver} on {d}. / Change approved by {approver} on {d}. / CAB approval recorded for {ref} on {d}.	Signed off by {approver}, dated {d}. / {approver} recorded an approval for this change on {d}. / Approval entry dated {d}, approver {approver}.
approval_assertion	not_recorded	Approval: none recorded. / Awaiting approval; no approver has signed off. / No CAB approval is on record for this change.	No sign-off has been entered for this change. / The approval log for this change is empty. / Nobody has approved this change yet.

field	value	validation bank	test bank
data_residency_region	us	Data residency: United States. / Customer data for this resource is stored in US data centres. / Residency agreement {doc} places the data in the United States.	Hosting region: US. / The data for this resource are kept in the United States. / Storage location per contract {doc}: United States.
data_residency_region	eu	Data residency: European Union. / Customer data for this resource is stored in EU data centres. / Residency agreement {doc} places the data in the European Union.	Hosting region: EU. / The data for this resource are kept in the European Union. / Storage location per contract {doc}: European Union.
data_residency_region	apac	Data residency: Asia-Pacific. / Customer data for this resource is stored in APAC data centres. / Residency agreement {doc} places the data in the Asia-Pacific region.	Hosting region: APAC. / The data for this resource are kept in the Asia-Pacific region. / Storage location per contract {doc}: Asia-Pacific.

E2 statement banks (CH-B1).

field	value	validation bank	test bank
approval_assertion	recorded	Approval: granted by {approver} on {date}. / Change approved by {approver} under {ref}. / CAB approval recorded for {ref} on {date}.	Signed off by {approver}, dated {date}. / {approver} recorded an approval for this change. / Approval entry dated {date}, approver {approver}.
approval_assertion	not_recorded	Approval: none recorded. / Awaiting approval; no approver has signed off. / No CAB approval is on record for this change.	No sign-off has been entered for this change. / The approval log for this change is empty. / Nobody has approved this change yet.
branch	main	Target branch: main. / The change merges into the main branch. / Branch under change: main (trunk).	Merge target: main. / This patch goes onto main. / Destination branch is main.
branch	staging	Target branch: staging. / The change merges into the staging branch. / Branch under change: staging.	Merge target: staging. / This patch goes onto the staging branch. / Destination branch is staging.
branch	feature	Target branch: a feature branch. / The change lands on a feature branch. / Branch under change: feature/{slug}.	Merge target: feature/{slug}. / This patch goes onto a feature branch. / Destination branch is a feature branch.

field	value	validation bank	test bank
environment	production	Target environment: production. / This change deploys to the production environment. / Environment: prod (customer-facing).	Deploys to: production. / Rollout target is the production environment. / Runs against prod.
environment	staging	Target environment: staging. / This change deploys to the staging environment. / Environment: pre-production staging.	Deploys to: staging. / Rollout target is the staging environment. / Runs against staging.
environment	development	Target environment: development. / This change deploys to the development environment. / Environment: dev sandbox.	Deploys to: development. / Rollout target is the development environment. / Runs against dev.

Admission thresholds, E1. Fitted on split A at the 1% ceiling. “Never true” and “never false” mean that no candidate threshold met the ceiling, so the value is never admitted in that direction.

field	value	Jev τ^{true}	Jev τ^{false}	Haiku τ^{true}	Haiku τ^{false}
approval_assertion	not_recorded	0.28	0.29	never true	never false
approval_assertion	recorded:2026-06-01	0.45	0.02	0.05	0.05
approval_assertion	recorded:2026-06-15	0.95	0.02	0.02	0.05
approval_assertion	recorded:2026-07-01	0.94	0.02	0.15	0.01
approval_assertion	recorded:2026-07-15	0.64	0.02	0.50	0.01
approval_assertion	recorded:2026-08-01	0.63	0.02	0.50	0.01
approval_assertion	recorded:2026-08-15	0.95	0.02	0.50	0.01
data_residency_region	apac	0.12	0.02	0.05	0.05
data_residency_region	eu	0.52	0.52	0.30	0.30
data_residency_region	us	0.35	0.02	0.50	0.00

Admission thresholds, E2.

field	value	Jev τ^{true}	Jev τ^{false}	Haiku τ^{true}	Haiku τ^{false}
approval_assertion	not_recorded	0.35	0.09	0.98	0.02
approval_assertion	recorded	0.42	0.46	0.30	0.30
branch	feature	0.57	0.57	0.34	0.05
branch	main	0.21	0.21	0.50	0.33
branch	staging	0.20	0.20	0.35	0.35
environment	development	0.28	0.28	1.00	0.34
environment	production	0.07	0.40	0.95	0.33
environment	staging	0.39	0.39	0.40	0.25

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